Sean Vincent Confoy

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EDUCATION

Vanderbilt University — **Nashville, Tennessee**
Bachelor of Engineering in Mechanical Engineering, Summa Cum Laude — August 2022 - May 2026
- **GPA**: **3.99/4.0** |Academic Achievement Award Recipient | Dean's List (8/8) | **SAT**: 1580

Relevant Coursework: Energetics | Mechanics of Materials | Linear Algebra | Aerodynamics | Circuits | Heat Transfer

Davidson Day School — **Cornelius, North Carolina**
High School Graduate — Graduated May 2022
Weighted GPA: 4.59 **Unweighted:** 4.0/4.0
Honors: Valedictorian | Harvard Book Award | Best Mathematician (2x) | Best Scientist (1x)

DESIGN ENGINEERING SKILLS

CAD/Tools: SolidWorks (CSWA), Fusion 360, Inventor, AutoCAD, Python, MATLAB, LabVIEW, Mathematica
Manufacturing: CNC Mill, Lathe, Welding, Soldering, Composite Manufacturing, Vacuum Forming, Bandsaw

PROFESSIONAL EXPERIENCE

The Boring Company — **Bastrop, Texas**
Mechanical Design Engineer (Associate) — June 2026– Present
- Designed a weldment shroud for a winch from concept to production, including SolidWorks models, fabrication drawings, weld drawings, and cut lists
- Designed a gantry-mounted control panel integrating pipe retraction, winch controls, emergency stop circuitry, and iPad mounting/charging to improve operator usability
- Designed and detailed a sheet metal storage assembly with gusset reinforcement for drills, couplers, isolation valves, and hardware, producing manufacturing-ready drawings
- Tested optimal pressurized-air operating parameters for grout and accelerant delivery through utility extension piping
- Engineered a hardline pressurized air delivery system and hose reel assembly, conducting pressure drop, flow rate, and weld strength calculations to ensure structural integrity and reliable operation during accelerant and grout flushing

Vanderbilt Aerospace Design Laboratory — **Nashville, Tennessee**
Lead Payload Engineer — August 2025 – May 2026
- Placed 2nd overall among 33 universities and won the NASA USLI Payload Award for payload design and performance
- Engineered retractable shovel payload with multi-stage gearbox, enabling reliable soil extraction for NASA competition rocket
- Led a team of 4 engineers through design reviews, testing, and risk analysis, ensuring deadlines were met under requirements
- Authored technical reports documenting failure analysis and design trade studies presented to faculty and industry reviewers

PAE Engineers — **New York, New York**
Mechanical Engineering Intern — May 2025 – August 2025
- Engineered HVAC duct layouts meeting ASHRAE airflow standards for high-profile technology clients like Amazon and Intuit
- Revised HVAC and plumbing systems based on client feedback, producing updated schematics and water riser diagrams
- Performed load calculations to determine optimal CFM and evaluated exhaust duct runs to determine static pressure losses

Huntington Ingalls Industries — **Newport News, Virginia**
Mechanical Engineering Intern — June 2024 – August 2024
- Developed 3D CAD models of naval infrastructure, including a 1.12 million square foot dry dock for housing aircraft carriers
- Produced numerous two-dimensional marine and naval drawing schematics used for renovating foundation and piping systems
- Conducted inspections aboard the USS John F. Kennedy aircraft carrier, evaluating mechanical, electrical, and piping systems

MEMBERSHIPS AND INTERESTS

Security Clearance: U.S. Secret (initiated via HII internship—favorably adjudicated in late 2024, still need to be read in)
Memberships and Interests: Tau Beta Pi Honor Society | Sigma Nu | Rock Climbing | Jiu Jitsu | Archaeology | Snowboarding